

Physical and Hydro Geochemical Impact Assessment of Inactive Mines in and Around Southwestern Part of Cuddapah Basin Using a Conceptual Site Model (CSM)



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Abstract Inactive mines have a tremendous influence on the environment and human health, and are frequently the result of a lack of attention and planning. As a result, this report outlines all of the inactive mine site characteristics in terms of physical and chemical consequences, community safety, and human and environmental health threats. To that purpose, this work develops a conceptual site model that may be used to locate, describe, and map inactive mining locations. Inactive mines are creating haphazard dangers to local communities and the environment in the research area, based on site characteristics such as vertical and horizontal openings, toxic impoundments, and steep waste dumps. Water chemistry studies have been undertaken near inactive mines in terms of chemical impact evaluation. Groundwater in the vicinity of inactive mining zones has a high pH and alkalinity, and the dissolved ion concentration (TDS) of 2% is more than the permitted limit, according to the analytical data. Sulfate and chloride are two anions that are within the acceptable range. The nitrate concentrations in the research area ranged from 0.14 to 589 mg/L, and anthropogenic input from inactive mining areas caused 45% of the groundwater samples to exceed the permissible range. As, Ni, Pb, and Sr all have greater heavy metal concentrations than the permissible limit. This suggests that the greater concentration of heavy metals in the research area is due to human input from inactive mine disposal sites. As a result, the rehabilitation program must be carried out in order to eliminate haphazard difficulties in the research region caused by inactive problems.

Keywords Conceptual site model (CSM) · Groundwater quality · Inactive mines · Southwestern part of Cuddapah basin

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